

Realtek Wi-Fi SDK for Android 10.x

ver. 1.0.0

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Release History

1.0.0	2019/10/14	1. First formal release
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Realtek

SDK packages

- hardware/realtek/*

Folder to store private code from Realtek.

Introduction

This document provides a simple guide to help engineers to apply Realtek Wi-Fi solution onto their Android 10.x system. The following two scenarios are supported

- **STA/AP** _ Switch between STA mode and AP mode
- **(STA+P2P)/AP** _ Switch between STA+P2P(Wi-Fi Direct) concurrent mode and AP mode

To port Realtek Wi-Fi driver onto Android 10.x platform, you can go through the following guide with reference codes within our driver package's [realtek_wifi_SDK_for_android_10.x_20191008.tgz](#).

Because Android's SDK may differ from platform to platform, our reference codes may not be applied on every platform without modifications. You should check if our reference code is suitable for you to use.

In this document, ANDROID_SDK is the path of top folder of the target Android SDK; this term is used in the following paragraphs.

1. Copy Necessary Files into SDK

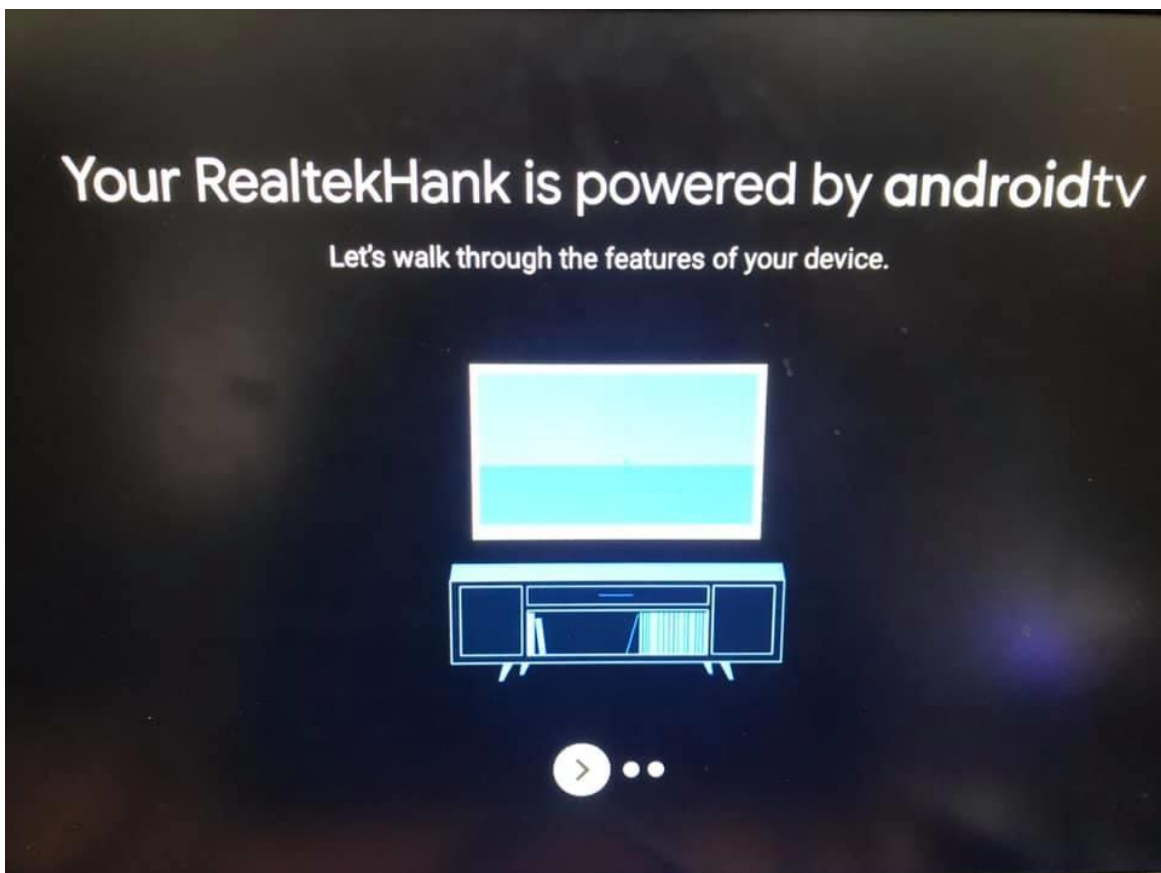
After unzipping [realtek_wifi_SDK_for_android_10.x_20191008.tgz](#), copy the following folder into ANDROID_SDK/hardware/folder:

- hardware/realtek

2. Platform Related Files

2.1. Suggested Platforms

Realtek Hank - DHC 1319 platform



2.2. BoardConfig.mk

To apply Realtek Wi-Fi solution onto your Android 10.x system, you need to define some compile-time variables in BoardConfig.mk of your platform. In general, the BoardConfig.mk file is located in:

ANDROID_SDK /device/<soc_vendor_name>/<board_name>/

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/BoardConfig.mk

```

BOARD_WIFI_VENDOR := realtek
ifeq ($(BOARD_WIFI_VENDOR), realtek)
    WPA_SUPPLICANT_VERSION := VER_0_8_X
    BOARD_WPA_SUPPLICANT_DRIVER := NL80211

    #CONFIG_DRIVER_WEXT := y
    BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl
    BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl
    BOARD_HOSTAPD_DRIVER := NL80211

    BOARD_WLAN_DEVICE := realtek
endif

```

- **BOARD_WIFI_VENDOR := realtek**

To distinguish the platform Wi-Fi device from products of other vendors, we define variable BOARD_WIFI_VENDOR as realtek. This is for compile-time choices to be applied for Realtek Wi-Fi solutions.

- **WPA_SUPPLICANT_VERSION := VER_0_8_X**

For Android 10, please set WPA_SUPPLICANT_VERSION as VER_0_8_X to use wpa_supplicant_8.

- **BOARD_WPA_SUPPLICANT_DRIVER := NL80211**

- **BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl**

- **BOARD_HOSTAPD_DRIVER := NL80211**

- **BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl**

We use NL80211 as the driver interface for wpa_supplicant and hostapd to communicate with driver and provide lib_driver_cmd_rtl as the private library.

- **BOARD_WLAN_DEVICE**

BOARD_WLAN_DEVICE is used to choose which vendor wifi_hal should be applied. You have to set **BOARD_WLAN_DEVICE := realtek** to use realtek's wifi_hal.

2.3. init.xxx.rc

For Wi-Fi to operate properly, we need some actions and daemons to be defined as service inside init.xxx.rc. In general, the init.xxx.rc file is located in:

ANDROID_SDK/device/<soc_vendor_name>/<board_name>/

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/common/init.hank.rc

Please add the service definitions below:

- **mkdir for wpa_supplicant and copy config file**

Please make sure the directories used by the Wireless and related wpa_supplicant config files will be created and copied in init rc.

```
on zygote-start
# Create the directories used by the Wireless subsystem
# Copy wpa_supplicant config files to related dir
mkdir /data/misc/wifi 0770 wifi wifi
mkdir /data/misc/wifi/wpa_supplicant 0770 wifi wifi
mkdir /data/vendor/wifi 0771 wifi wifi
mkdir /data/vendor/wifi/wpa 0770 wifi wifi
mkdir /data/vendor/wifi/wpa/sockets 0770 wifi wifi
mkdir /data/misc/dhcp 0770 dhcp dhcp
chown dhcp dhcp /data/misc/dhcp
```

- **insmod**

Please select one of action definitions below according to your requirement.

Meanwhile, please make sure your wlan.ko has right seclabel and mode as

```
-rw-r--r-- 1 root root u:object_r:vendor_file:s0 3429448 1970-01-01 00:19 wlan.ko
```

You can use command “ls -alZ” to check seclabel of wlan.ko.

(For concurrent mode)

```
on boot

exec u:r:vendor_modprobe:s0 root root -- /vendor/bin/toybox_vendor insmod
/system/vendor/lib/modules/wlan.ko ifname=wlan0 if2name=p2p0
```

(For STA only)

on boot

```
exec u:r:vendor_modprobe:s0 root root -- /vendor/bin/toybox_vendor insmod  
/system/vendor/lib/modules/wlan.ko ifname=wlan0
```

- **wpa_supplicant**

Please define wpa_supplicant service as below.

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/common/prebuilt/vendor/etc/init/wpa_
supplicant.rc

```
service wpa_supplicant /vendor/bin/hw/wpa_supplicant \  
-O/data/vendor/wifi/wpa/sockets \  
-g@android:wpa_wlan0  
interface android.hardware.wifi.supplicant@1.0::ISupplicant default  
interface android.hardware.wifi.supplicant@1.1::ISupplicant default  
interface android.hardware.wifi.supplicant@1.2::ISupplicant default  
socket wpa_wlan0 dgram 660 wifi wifi  
class main  
disabled  
oneshot
```

2.4. Others

For topics mentioned here, you can add the following code segments in any .mk file which your platform will use.

- **Add wifi related packages**

These packages are needed for Wifi support in Android 10.x, please make sure these packages are added in .mk

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/device.mk

```
PRODUCT_PACKAGES += \
libwpa_client wpa_supplicant hostapd wificond wifilogd wpa_supplicant.conf
hostapd.conf libwifi-hal android.hardware.wifi.suplicant@1.0-service
android.hardware.wifi.suplicant@1.1-service
android.hardware.wifi.suplicant@1.2-service android.hardware.wifi@1.0-
service android.hardware.wifi@1.0-service-lib android.hardware.wifi.hostapd@1.0-
service
```

- **Add android.hardware.wifi.xml**

To claim Wi-Fi support for your device, please add the rule in the PRODUCT_COPY_FILES variable to copy the permission definition file of Wi-Fi to the /system/etc/permissions/ folder of your system image.

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

```
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.wifi.xml:$(TARGET_COPY_OUT_VENDOR)/etc/permissions/android.hardware.wi
fi.xml
```

- **Add android.hardware.wifi.direct.xml**

To claim Wi-Fi Direct (P2P) support for your device, please add the rule in the PRODUCT_COPY_FILES variable to copy the permission definition file of Wi-Fi Direct to the /system/etc/permissions/ folder of your system image.

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

```
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.wifi.direct.xml:$(TARGET_COPY_OUT_VENDOR)/etc/permissions/android.hardware.wifi.direct.xml
```

Make sure your driver is configured for STA+P2P concurrent mode or you may encounter error when you open the Wi-Fi. Please refer to “**5. Driver Configurations for Android 10.x**”

If we use Realtek DHC 1319 platform as example, it is located at
 ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

- **Set wifi.interface**

To specify the wifi interface name in Android, a system property named `wifi.interface` is needed. For Realtek Wi-Fi driver, Wi-Fi interface name is assigned with `wlan0`. In general, the default `wifi.interface` is `wlan0`.

If we use Realtek DHC 1319 platform as example, it is located at
 ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

```
PRODUCT_PROPERTY_OVERRIDES += \
    wifi.interface=wlan0
```

- **Set wifi.direct.interface**

If you require p2p support, you have to set `wifi.direct.interface` as `p2p0`

If we use Realtek DHC 1319 platform as example, it is located at
 ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

```
PRODUCT_PROPERTY_OVERRIDES += \
    wifi.direct.interface=p2p0
```

- **Toybox_vendor**

In android 10.x, we need `toybox_vendor` to `insmod`. You can set the config in `device.mk`
`PRODUCT_SHIPPING_API_LEVEL := 26` (more than or equal with 26)

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/device/realtek/hank/common/deviceCommon.mk

And `PRODUCT_SHIPPING_API_LEVEL := 29`

You should modify `$ANDROID_SDK/external/toybox/Android.mk` as below to build

toybox_vendor, if the config is not set.

```
#ifeq ($(PRODUCT_FULL_TREBLE),true)
#####
# static version to be installed in /vendor
#
.
LOCAL_MODULE := toybox_vendor
.
include $(BUILD_EXECUTABLE)
#endif
```

If we use Realtek DHC 1319 platform as example, it is located at
ANDROID_SDK/frameworks/opt/net/wifi/libwifi_hal/Android.mk

```
# Pick a vendor provided HAL implementation library.
# =====
LIB_WIFI_HAL := libwifi-hal-fallback
.
else ifeq ($(BOARD_WLAN_DEVICE), realtek)
# support RTK WIFI HAL
LIB_WIFI_HAL := libwifi-hal-rtk
endif
.
```

● Device manifest file

The Device manifest file is provided by the device. It lives in the Android source tree at device/\${VENDOR}/\${DEVICE}/manifest.xml and on the device at /vendor/manifest.xml

If we use Realtek DHC 1319 platform as example, it is located at

ANDROID_SDK/frameworks/opt/net/wifi/libwifi_hal/Android.mk

RTK wifi needs **Iwifi** , **ISupplicant** and **IHostapd**. Please check device/\${VENDOR}/\${DEVICE}/manifest.xml,
It should have IWifi and ISupplicant description as below.

```

<hal format="hidl">
  <name>android.hardware.wifi</name>
  <transport>hwbinder</transport>
  <version>1.3</version>
  <interface>
    <name>IWifi</name>
    <instance>default</instance>
  </interface>
</hal>
<hal format="hidl">
  <name>android.hardware.wifi.suplicant</name>
  <transport>hwbinder</transport>
  <version>1.2</version>
  <interface>
    <name>ISupplicant</name>
    <instance>default</instance>
  </interface>
</hal>
<hal format="hidl">
  <name>android.hardware.wifi.hostapd</name>
  <transport>hwbinder</transport>
  <version>1.1</version>
  <interface>
    <name>IHostapd</name>
    <instance>default</instance>
  </interface>
</hal>

```

3. System Resource Configurations

You should set the following four resource configurations for your platform to configure the network function and enable the corresponding UI interface. In general, you can set the following configurations in your platform dependent config.xml file.

If we use Realtek DHC 1319 platform as example, it is located at
 ANDROID_SDK/frameworks/base/core/res/res/values/config.xml

Or the global config.xml file:

ANDROID_SDK/frameworks/base/core/res/res/values/config.xml

● networkAttributes

To define the available network interfaces, make sure the wifi interface items is defined in the networkAttributes resource configuration in the config.xml.

For example:

```

<string-array translatable="false" name="networkAttributes">
  <item>"wifi,1,1,1,-1,true"</item>
  <item>"bluetooth,7,7,0,-1,true"</item>
  <item>"ethernet,9,9,2,-1,true"</item>
</string-array>

```

- **radioAttributes**

To define the available network resources, we need to define network items for wifi in the radioAttributes resource configuration. For example:

```
<string-array translatable="false" name="radioAttributes">
  <item>"1,1"</item>
  <item>"7,1"</item>
  <item>"9,1"</item>
</string-array>
```

- **config_tether_wifi_regexs**

The interfaces set here are tetherable Wi-Fi interfaces which will be used as interfaces for Wi-Fi LAN port. We use 'wlan0' by default when our Wi-Fi is set as softap mode. So it needs to set 'wlan0' here. For example:

```
<string-array translatable="false" name="config_tether_wifi_regexs">
  <item>"wlan0"</item>
</string-array>
```

- **config_tether_upstream_types**

The connection types set here are used as the interfaces for WAN port to connect to internet. For example, adding Wi-Fi and Ethernet:

```
<integer-array translatable="false" name="config_tether_upstream_types">
  <item>1</item>
  <item>9</item>
</integer-array>
```

At least one item should be declared here to enable the Tethering & portable hotspot. Wireless & portable hotspot.

To know the definition and set other upstream connection types, please refer to [ANDROID_SDK/frameworks/base/core/java/android/net/ConnectivityManager.java](#).

- **config_enableWifiDisplay**

To enable Wi-Fi Display(Miracast) function, set config_enableWifiDisplay to true:

```
<bool name="config_enableWifiDisplay">true</bool>
```

4. wpa_supplicant_8

We provide **wpa_supplicant_8_10.x_rtw_29226.20191002** or **newer** version in this release package. You can:

- Use the **wpa_supplicant_8_10.x_rtw_29226.20191002** instead of the original
 1. Backup and remove the original external/wpa_supplicant_8/ folder
 2. Extract and copy the **wpa_supplicant_8_10.x_rtw_xxxx** tar file to the external/ folder of your Android SDK.
 3. Rename **wpa_supplicant_8_10.x_rtw_xxxx** as **wpa_supplicant_8**.

5. Driver Configurations for Android 10.x

Android 10.x support two scenarios for Wi-Fi solution:

- **STA/AP – Switch between STA and AP mode**
- **(STA+P2P)/AP – Switch between STA+P2P concurrent and AP mode**

The configuration of driver to fit the requirement of each scenario, see the following table:

MACRO	STA /AP	(STA+P2P)/AP	Kernel ver.
CONFIG_IOCTL_CFG80211	Defined	Defined	ver. >= 2.6.35
RTW_USE_CFG80211_STA_EVENT	Defined	Defined	ver. >= 3.2.0
CONFIG_RADIO_WORK	Defined	Defined	-
CONFIG_CONCURRENT_MODE	Undefined	Defined	-
RTW_ENABLE_WIFI_CONTROL_FUNC	Defined for platform device/driver mechanism		
CONFIG_RTW_WIFI_HAL	Defined if android version >= 8.x		ver. >= 3.18

- **CONFIG_IOCTL_CFG80211** is used for driver to enable cfg80211 ioctl interface, which is required by Realtek Wi-Fi to operate on Android 10.x system.

- **RTW_USE_CFG80211_STA_EVENT** is used for driver to indicate new cfg80211 STA event, which is required by wpa_supplicant_8 Linux kernel supports this feature after kernel 3.2.
- **CONFIG_RADIO_WORK** is used for driver to enable 'radio work' mechanism in wpa_supplicant. In this MACRO definition, please refer to the driver's source code, please contact with Realtek technical windows for suitable driver.
- **CONFIG_CONCURRENT_MODE** is used for driver to enable concurrent mode, which is required by STA+P2P concurrent mode.
- **RTW_ENABLE_WIFI_CONTROL_FUNC** is used to register platform driver callbacks. If your platform needs those callbacks, please define this macro to register platform driver callback functions. For example, these functions include:

```
static struct platform_driver wifi_device = {
    .probe      = wifi_probe,
    .remove     = wifi_remove,
```

By default, the probe callback is used to set up Wi-Fi power and remove callback is used to close Wi-Fi power.

To compile Realtek Wi-Fi driver with the above setting, please refer to the following document:

document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf
Adding platform selection and setting sections for compilation settings of your platform.

For example, if you want to configure Realtek Wi-Fi driver for the (STA+P2P)/AP scenario, make sure the macros: CONFIG_IOCTL_CFG80211, RTW_USE_CFG80211_STA_EVENT, **CONFIG_RADIO_WORK** and CONFIG_CONCURRENT_MODE are defined into the EXTRA_CFLAGS settings as following:

```
CONFIG_PLATFORM_ANDROID_M60_SAMPLE = y
```

```
ifeq ($(CONFIG_PLATFORM_ANDROID_ML0_SAMPLE), y)
```

```
EXTRA_CFLAGS += -DCONFIG_LITTLE_ENDIAN
```

```
EXTRA_CFLAGS += -DCONFIG_CONCURRENT_MODE
```

```
EXTRA_CFLAGS += -DCONFIG_IOCTL_CFG80211 -DRTW_USE_CFG80211_STA_EVENT
```

```
EXTRA_CFLAGS += -DCONFIG_RADIO_WORK
```

```
ARCH := arm
```

```
CROSS_COMPILE := /toolchain/bin/arm-none-linux-gnueabi-
```

```
KSRC := /android_sdk/android_l/ kernel
```

```
endif
```

- **CONFIG_RTW_WIFI_HAL** is defined if android version is $\geq 8.x$:
For supporting Android version $\geq 8.x$, make sure **CONFIG_RTW_WIFI_HAL** is set to `y` in **Makefile** as follows,

```
CONFIG_RTW_WIFI_HAL = y
```

According to Google's suggestion, **you must use kernel 3.18 or newer.**

For more detail, you can refer

<https://source.android.com/devices/architecture/kernel/modular-kernels>

5.1 CONFIG_RTW_ANDROID

From Wifi driver version 5.9, a new setting **CONFIG_RTW_ANDROID** is added in Makefile. We can set **CONFIG_RTW_ANDROID** with the Android version in Makefile. e.g. **CONFIG_RTW_ANDROID = 10**

Please note that we must set **CONFIG_RTW_ANDROID** with correct Android version from wifi driver version 5.9, otherwise there will be problem in wifi driver for Android. And the default value of **CONFIG_RTW_ANDROID** is 0, which means the driver is for pure linux, not Android.

(**CONFIG_RTW_ANDROID=4** means Android 4.4)

Example in Makefile:

```
##### Android #####
# CONFIG_RTW_ANDROID - 0: no Android, 4/5/6/7/8/9/10 : Android version
CONFIG_RTW_ANDROID = 10

ifeq ($(shell test $(CONFIG_RTW_ANDROID) -gt 0; echo $$?), 0)
EXTRA_CFLAGS += -DCONFIG_RTW_ANDROID=$(CONFIG_RTW_ANDROID)
endif
```

Then most of the settings mentioned above are set automatically by Android version (CONFIG_RTW_ANDROID). We have seen in Makefile; Except CONFIG_CONCURRENT_MODE and RTW_ENABLE_WIFI_CONTROL_FUNC still need to be set manually, depends on the platforms in Makefile as above as before.

Drv_conf.h

```
#define CONFIG_IOCTL_CFG80211

#define RTW_USE_CFG80211_STA_EVENT

#if (CONFIG_RTW_ANDROID > 4)
#ifndef CONFIG_RADIO_WORK
#define CONFIG_RADIO_WORK
#endif
#endif

#if (CONFIG_RTW_ANDROID >= 8)
    #if (LINUX_VERSION_CODE >= KERNEL_VERSION(3,18,0))
    #ifndef CONFIG_RTW_WIFI_HAL
    #define CONFIG_RTW_WIFI_HAL
    #endif
    #else
    #error "Linux kernel version is too old\n"
    #endif
#endif
```

5.2 RTW_SINGLE_WIPHY

From driver v5.9, single wiphy is enabled as default. We can change back to the previous setting as before by mark off the line in include/drv_conf.h

```
/* Default enable single wiphy if driver ver >= 5.9 */
#define RTW_SINGLE_WIPHY
```

6. FAQ

6.1 Wi-Fi (STA mode)

6.1.1 Why Wi-Fi can't enable?

eck

The whole Wi-Fi enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface(s) created?**
 - insmod driver success
 - Wi-Fi device is recognized
- **Does wpa_supplicant run successfully?**
 - wpa_supplicant.conf (and p2p_supplicant.conf) exists and is correct
 - Service definition of wpa_supplicant exists and is correct
 - Binary file wpa_supplicant exists and is executable
- **Do connections of communication socket setup?**
 - Make sure the communication socket settings is matched below:
 - ◆ ctrl_interface in:
/data/vendor/wifi/wpa/wpa_supplicant.conf
(and /data/vendor/wifi/wpa/p2p_supplicant.conf)
 - ◆ Service definition of wpa_supplicant
 - ◆ Paths of communication socket in wifi.c

6.2 Portable Wi-Fi hotspot (AP mode)

6.2.1 Why Portable Wi-Fi hotspot can't enable?

The whole Portable Wi-Fi hotspot enabling procedure includes the following three main check points. Please check in sequence:

- **Is network interface created?**
 - insmod driver success
 - Wi-Fi device is recognized
 - wlan0 is created
- **Does netd and hostapd run successfully?**
 - /data/misc/wifi/hostapd.conf exists and is correct
 - Binary file netd and hostapd exist and are executable
- **Does dnsmasq run successfully?**
 - Binary file dnsmasq exist and are executable

6.3 Wi-Fi Direct (P2P mode)

6.3.1 There is no Wi-Fi Direct UI shown?

Please refer **Add android.hardware.wifi.direct.xml** in chapter 2.3. Others to enable Wi-Fi Direct functionality of Android P.

6.3.2 Wi-Fi Direct can't scan any peer?

First, make sure you have workable Wi-Fi Direct device nearby. Make them into Wi-Fi Direct device and wait for a while.

If there is still no peer shown the problem is usually caused by wrong service definition of wpa_supplicant. Please refer **wpa_supplicant** in chapter 2.2. **init.xxx.rc** to check your service definition of wpa_supplicant.

6.4 VTS test

6.4.1 VtsHalWifiSupplicantV1_0Host – ACS relative items failed

We have relative patches at hostapd. Please refer to 4. wpa_supplicant_8 to patch it..

6.5 CTS test

6.5.1 CtsNetTestCases – TCP keep alive item failed

We can disable `config_networkSupportedKeepaliveCount` in Android config.xml as below

```
<!-- Default supported concurrent socket keepalive slots per transport type, used by
ConnectivityManager.createSocketKeepalive() for calculating the number of keepalive
offload slots that should be reserved for privileged access. This string array should be
overridden by the device to present the capability of creating socket keepalives. -->

<!-- An Array of "[NetworkCapabilities.TRANSPORT_*],[supported keepalives] -->
<string-array translatable="false" name="config_networkSupportedKeepaliveCount">
```